Multi-band GNSS / NavIC Active Antenna - RF Front End

Design brief: architecture, simulated performance, filter rejection, interference (L1, L2/L5, NavIC S)

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RFM / Antenna Group, CRL-BG

Item	Entry
Prepared by	RFM / Antenna Group, CRL-BG
Prepared / Checked / Approved	_____ / _____ / _____
Basis	Manufacturer data sheets + circuit simulation (no hardware measurement yet)
Distribution	BEL internal only

Document control.

1 Key numbers

Band	MHz	Gain dB	NF dB	VSWR in	VSWR out	Ripple dB	IIP3 dBm	OP1dB dBm	C/N0 dB-Hz
L1 / E1 / B1C	1575.42	38.6	0.74	2.29	1.47	0.34	-7.9	16.8	44.8
B1I	1561.10	37.7	0.74	1.49	1.35	0.51	-	-	-
G1	1602.00	37.6	0.73	2.07	1.26	0.14	-	-	-
L5 / E5a / B2a	1176.45	42.0	0.59	1.63	1.42	1.14	-11.6	17.4	45.5
E5b / B2b	1207.14	41.6	0.60	2.75	2.17	0.42	-	-	-
L2	1227.60	40.1	0.62	1.96	1.58	0.41	-10.5	16.5	43.4
NavIC S	2492.03	35.0	1.17	1.24	1.15	0.08	-1.4	18.1	40.5

Table 1 - Simulated performance at 25 C (antenna feed RHCP -> RF output, 50 ohm).

Parameter	L-band	NavIC S
Stability	unconditional (mu >= 1.035)	unconditional (mu >= 1.053)
Reverse isolation	65 / 67 dB	58 dB
LHCP rejection / AR (hybrid)	31.8 dB / 0.45 dB	32.3 dB / 0.42 dB
Output noise power	-51.3 dBm	-59.7 dBm
Worst-case gain (data-sheet limits)	33 - 45 dB	-
DC	4 x TQP3M9037, 70 mA @ 5 V	2 x TQP3M9037, 70 mA @ 5 V
Total DC	420 mA @ +5 V = 2.1 W	

Table 2 - System parameters.

2 Architecture

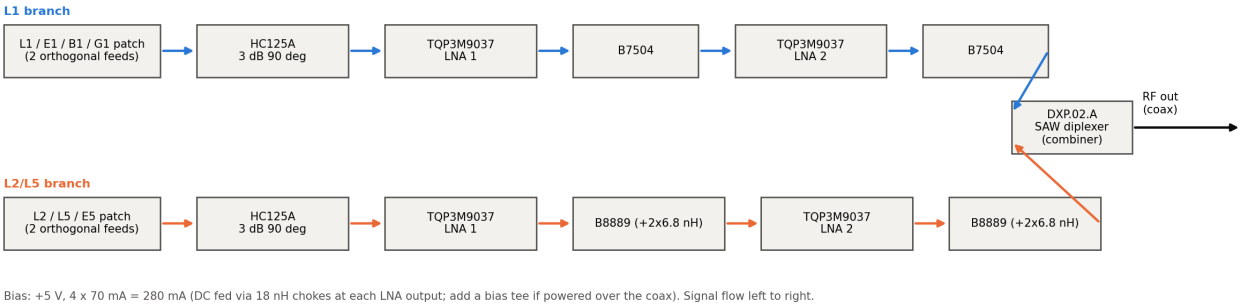


Figure 1 - L1 and L2/L5 chains, combined in the DXP.02.A.

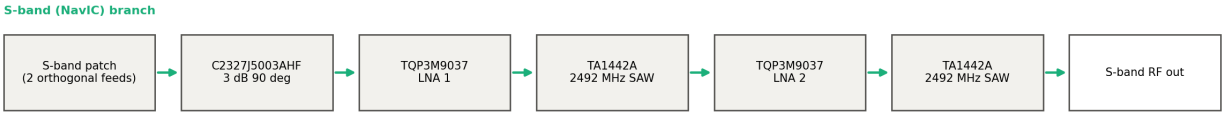


Figure 2 - NavIC S-band chain (separate output).

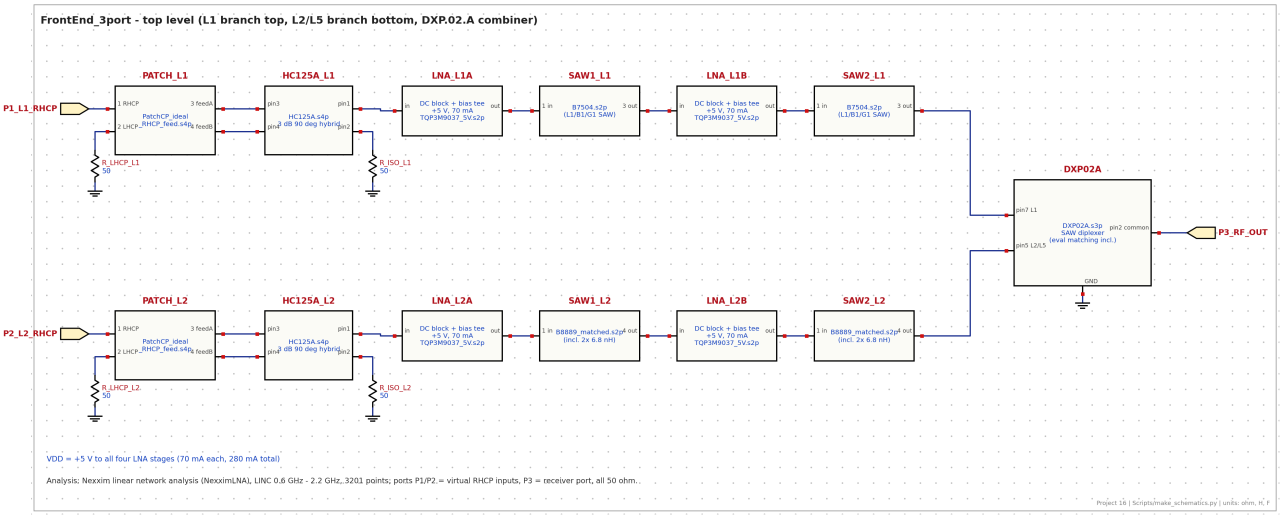


Figure 3 - L-band simulation schematic.

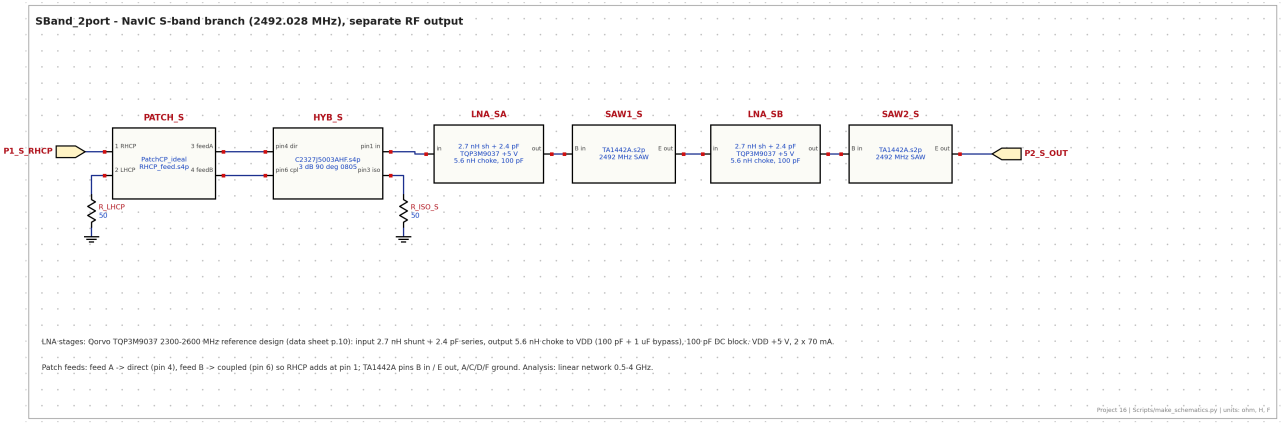


Figure 4 - S-band simulation schematic.

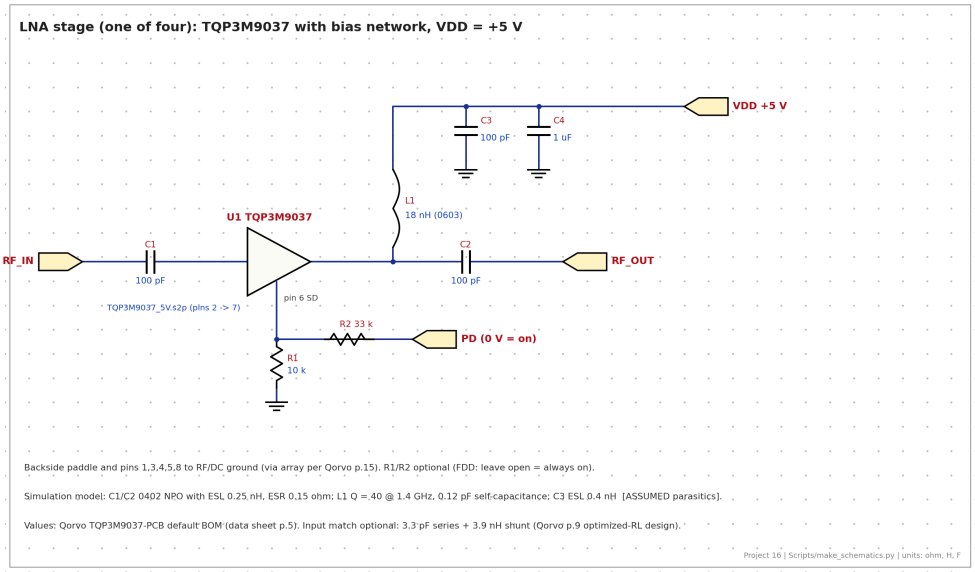
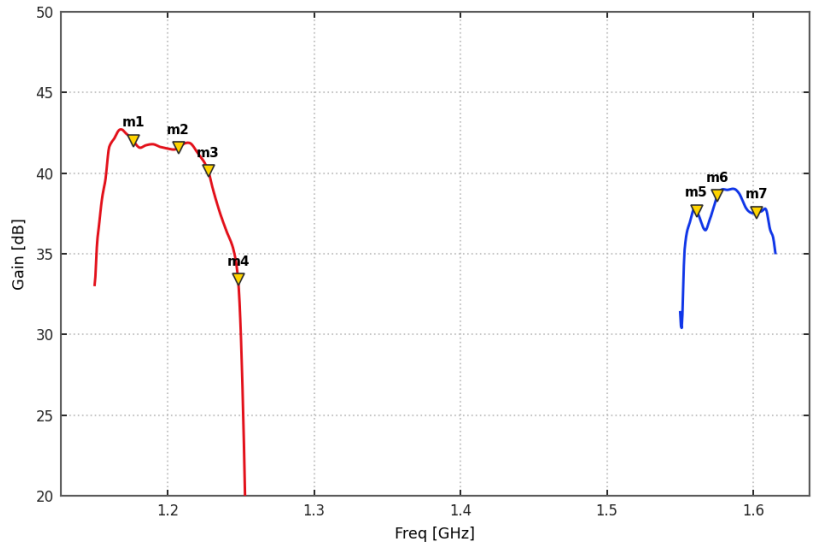


Figure 5 - LNA stage and bias network.

3 L-band results

XY Plot 4 - Gain in the GNSS bands *FrontEnd_3port*



Curve Info

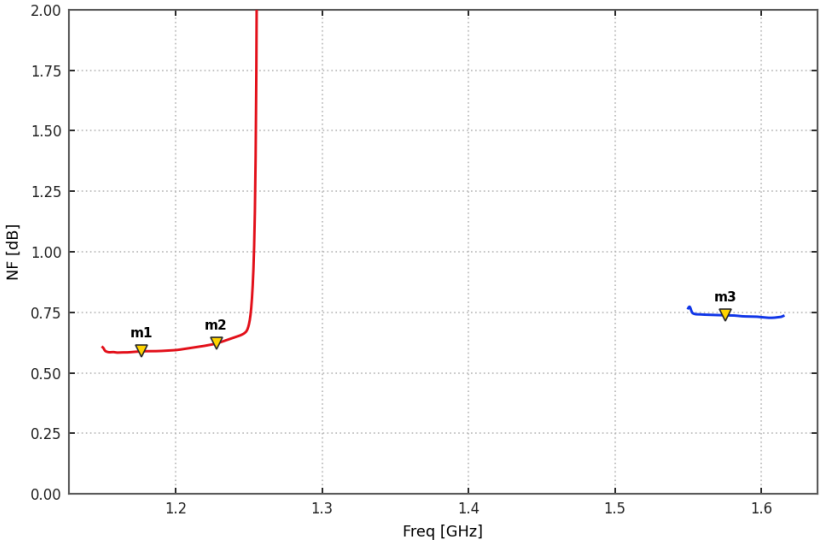
dB(S(P3_RF_OUT,P2_L2L5_RHCP))
SP_0p6_2p2GHz : Sweep

dB(S(P3_RF_OUT,P1_L1_RHCP))
SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.1764	41.991	S32
m2	1.2071	41.562	S32
m3	1.2276	40.115	S32
m4	1.2480	33.403	S32
m5	1.5611	37.672	S31
m6	1.5754	38.582	S31
m7	1.6020	37.564	S31

Figure - Gain.

XY Plot 5 - Noise figure *FrontEnd_3port*



Curve Info

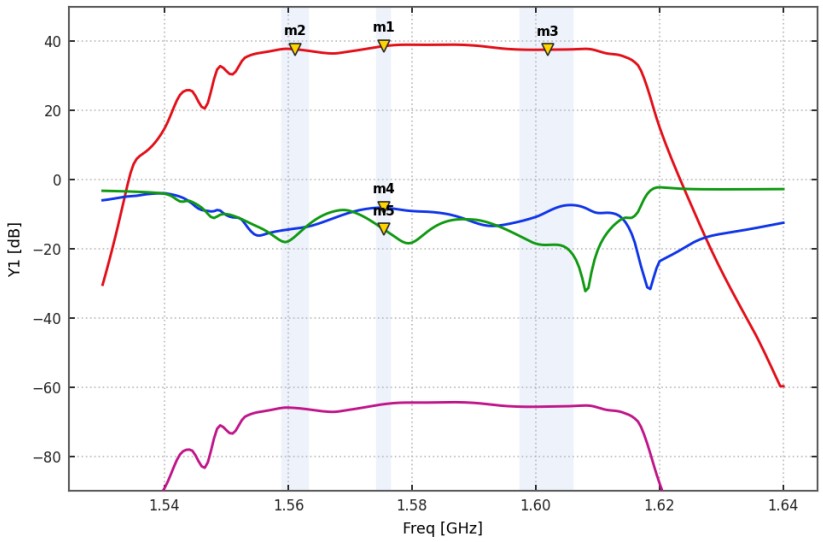
NF(P3_RF_OUT,P2_L2L5_RHCP)
SP_0p6_2p2GHz : Sweep

NF(P3_RF_OUT,P1_L1_RHCP)
SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.1764	0.589	NF L2/L5
m2	1.2276	0.621	NF L2/L5
m3	1.5754	0.738	NF L1

Figure - Noise figure.

XY Plot 1 - L1 branch S-parameters *FrontEnd_3port*



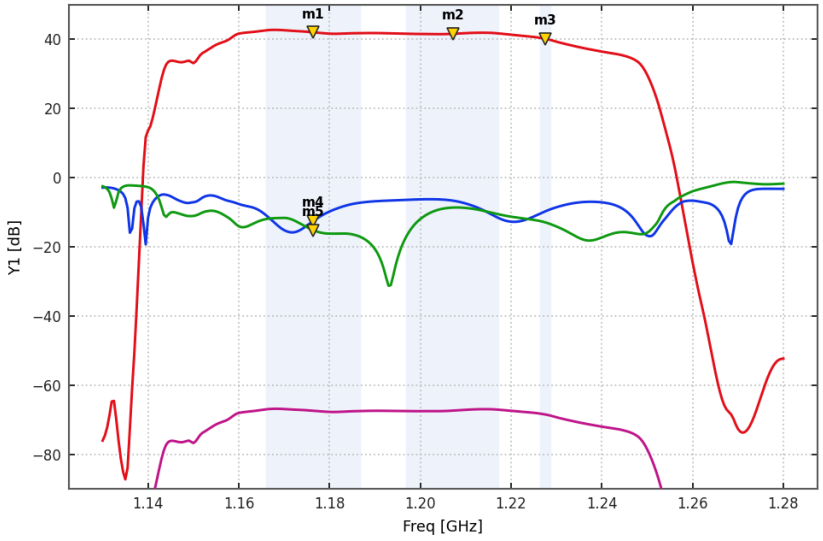
Curve Info

- dB(S(P3_RF_OUT,P1_L1_RHCP)) — SP_0p6_2p2GHz : Sweep
- dB(S(P1_L1_RHCP,P1_L1_RHCP)) — SP_0p6_2p2GHz : Sweep
- dB(S(P3_RF_OUT,P3_RF_OUT)) — SP_0p6_2p2GHz : Sweep
- dB(S(P1_L1_RHCP,P3_RF_OUT)) — SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.5754	38.582	S31
m2	1.5611	37.672	S31
m3	1.6020	37.564	S31
m4	1.5754	-8.139	S11
m5	1.5754	-14.349	S33

Figure - L1 branch S-parameters.

XY Plot 2 - L2/L5 branch S-parameters *FrontEnd_3port*



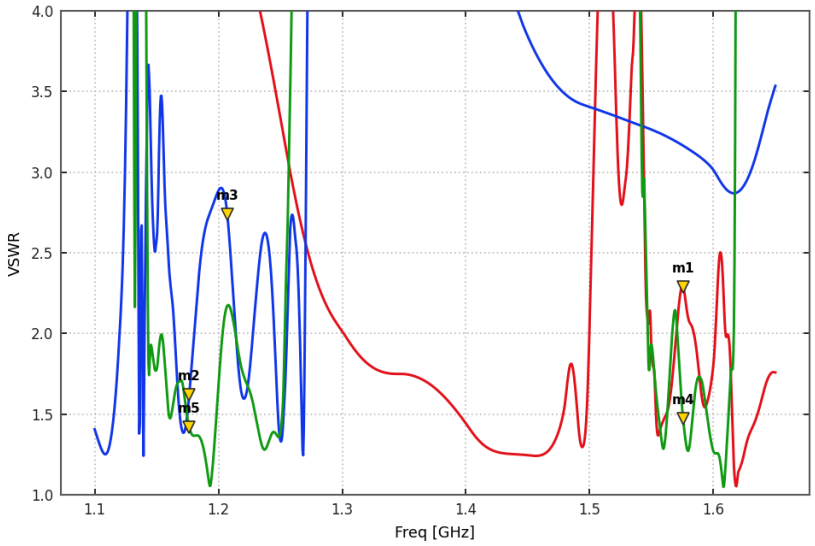
Curve Info

- dB(S(P3_RF_OUT,P2_L2L5_RHCP)) — SP_0p6_2p2GHz : Sweep
- dB(S(P2_L2L5_RHCP,P2_L2L5_RHCP)) — SP_0p6_2p2GHz : Sweep
- dB(S(P3_RF_OUT,P3_RF_OUT)) — SP_0p6_2p2GHz : Sweep
- dB(S(P2_L2L5_RHCP,P3_RF_OUT)) — SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.1764	41.991	S32
m2	1.2071	41.562	S32
m3	1.2276	40.115	S32
m4	1.1764	-12.498	S22
m5	1.1764	-15.216	S33

Figure - L2/L5 branch S-parameters.

XY Plot 6 - VSWR *FrontEnd_3port*



Curve Info

VSWR(P1_L1_RHCP)
SP_0p6_2p2GHz : Sweep

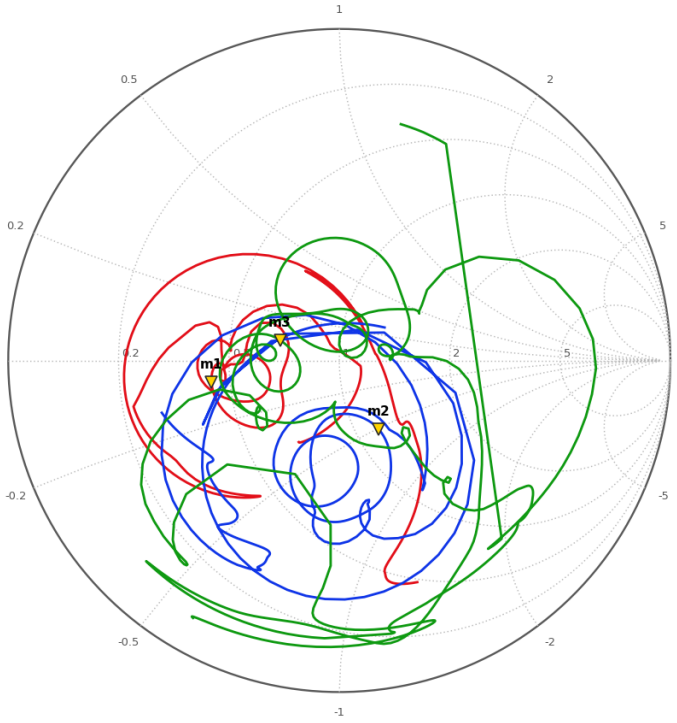
VSWR(P2_L2L5_RHCP)
SP_0p6_2p2GHz : Sweep

VSWR(P3_RF_OUT)
SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.5754	2.288	P1
m2	1.1764	1.622	P2
m3	1.2071	2.740	P2
m4	1.5754	1.474	P3
m5	1.1764	1.420	P3

Figure - VSWR.

Smith Plot 1 - Port reflections *FrontEnd_3port*



Curve Info

S(P1_L1_RHCP,P1_L1_RHCP)
SP_0p6_2p2GHz : Sweep

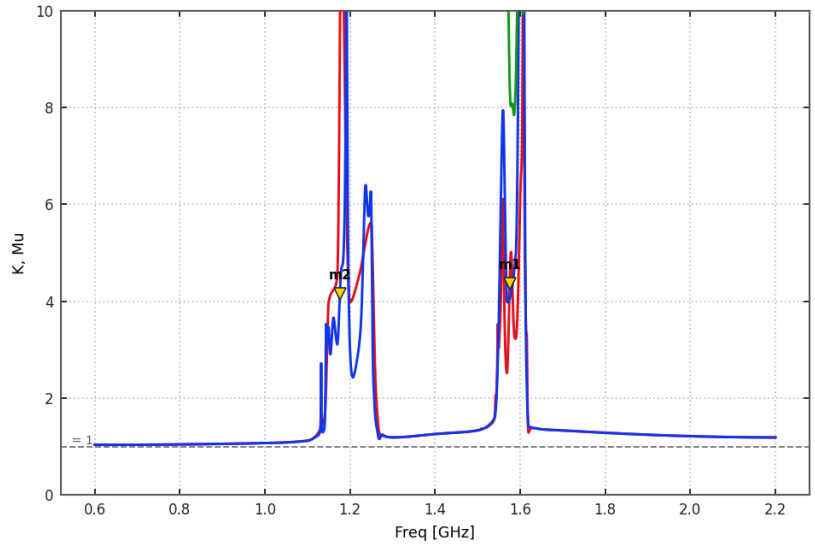
S(P2_L2L5_RHCP,P2_L2L5_RHCP)
SP_0p6_2p2GHz : Sweep

S(P3_RF_OUT,P3_RF_OUT)
SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.5754	22.0-j3.4	S11
m2	1.1764	57.5-j25.2	S22
m3	1.5754	34.6+j4.4	S33

Figure - Port reflections.

XY Plot 7 - Stability *FrontEnd_3port*



Curve Info

Mu1 (L1 branch)
SP_0p6_2p2GHz : Sweep

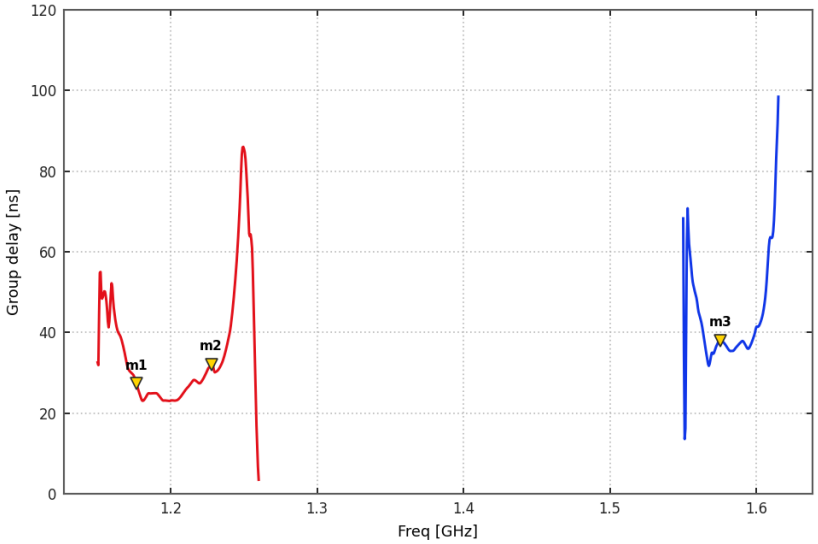
Mu1 (L2/L5 branch)
SP_0p6_2p2GHz : Sweep

StabilityFactor K (L1 branch)
SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.5754	4.370	Mu L1
m2	1.1764	4.155	Mu L2/L5

Figure - Stability.

XY Plot 8 - Group delay *FrontEnd_3port*



Curve Info

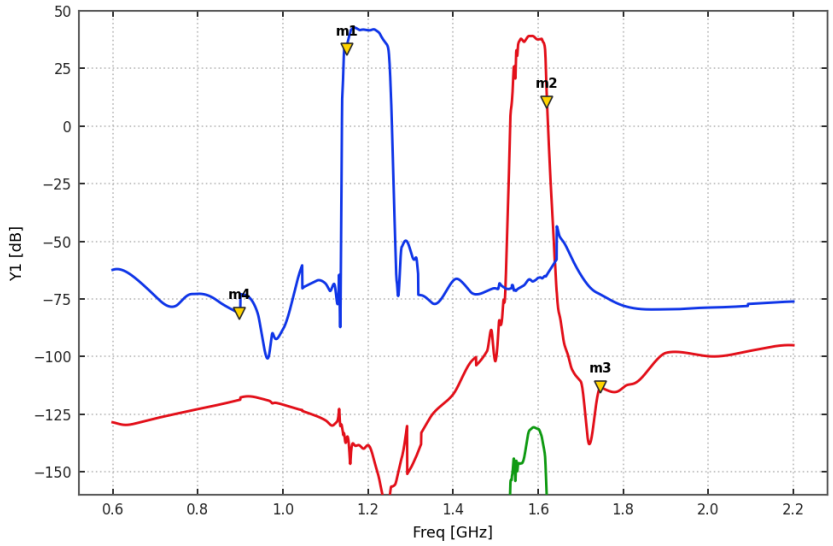
GroupDelay(P3_RF_OUT,P2_L2L5_RHCP) [ns]
SP_0p6_2p2GHz : Sweep

GroupDelay(P3_RF_OUT,P1_L1_RHCP) [ns]
SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.1764	27.353	GD L2/L5
m2	1.2276	32.102	GD L2/L5
m3	1.5754	38.021	GD L1

Figure - Group delay (estimate).

XY Plot 3 - Wide-band transmission *FrontEnd_3port*



Curve Info

dB(S(P3_RF_OUT,P1_L1_RHCP))
SP_0p6_2p2GHz : Sweep

dB(S(P3_RF_OUT,P2_L2L5_RHCP))
SP_0p6_2p2GHz : Sweep

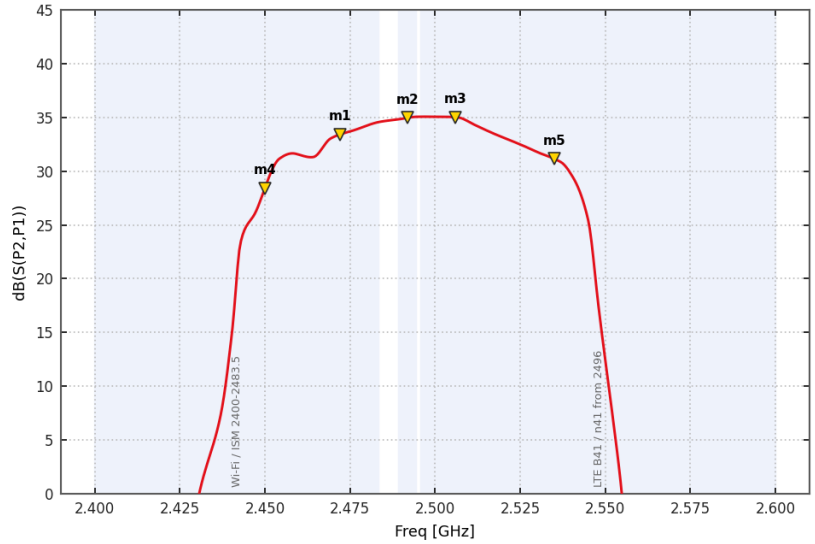
dB(S(P2_L2L5_RHCP,P1_L1_RHCP))
SP_0p6_2p2GHz : Sweep

	Freq [GHz]	Value	Trace
m1	1.1500	33.074	S32
m2	1.6210	10.249	S31
m3	1.7475	-113.117	S31
m4	0.8975	-81.234	S32

Figure - Wide-band transmission.

4 NavIC S-band results

XY Plot 2 - Gain, NavIC S band *SBand_2port*



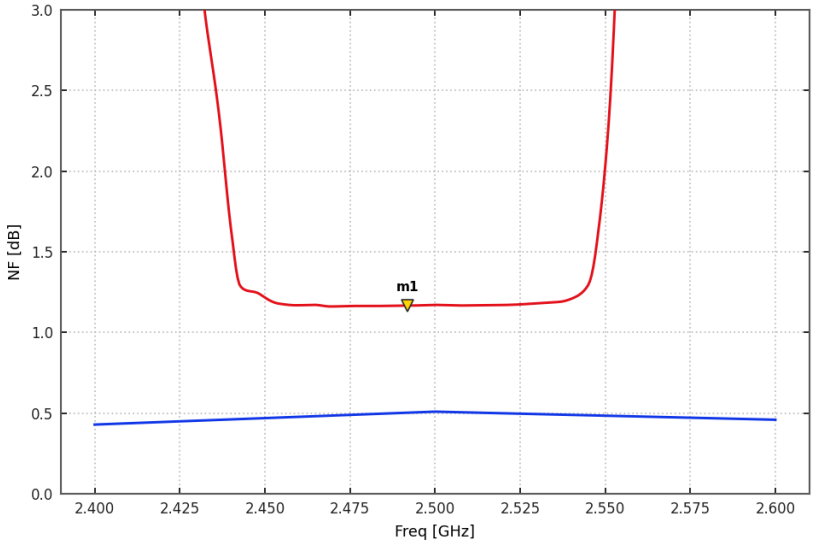
Curve Info

dB(S(P2_S_OUT,P1_S_RHCP))
SP_SBand : Sweep

	Freq [GHz]	Value	Trace
m1	2.4720	33.400	S21
m2	2.4920	34.954	S21
m3	2.5060	35.010	S21
m4	2.4500	28.400	S21
m5	2.5350	31.152	S21

Figure - Gain.

XY Plot 5 - Noise figure *SBand_2port*



Curve Info

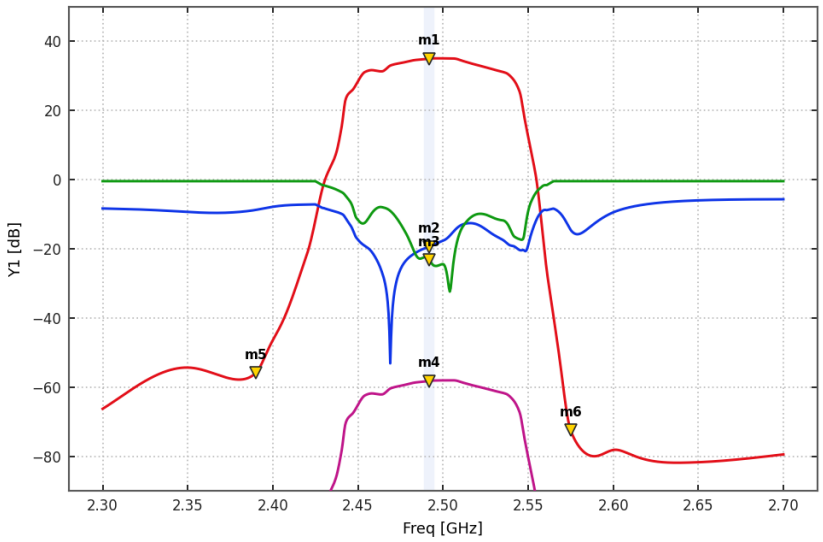
NF(P2_S_OUT,P1_S_RHCP)
SP_SBand : Sweep

NFmin TQP3M9037 (data sheet)
table, 5 V

	Freq [GHz]	Value	Trace
m1	2.4920	1.167	NF

Figure - Noise figure.

XY Plot 1 - S-parameters *SBand_2port*



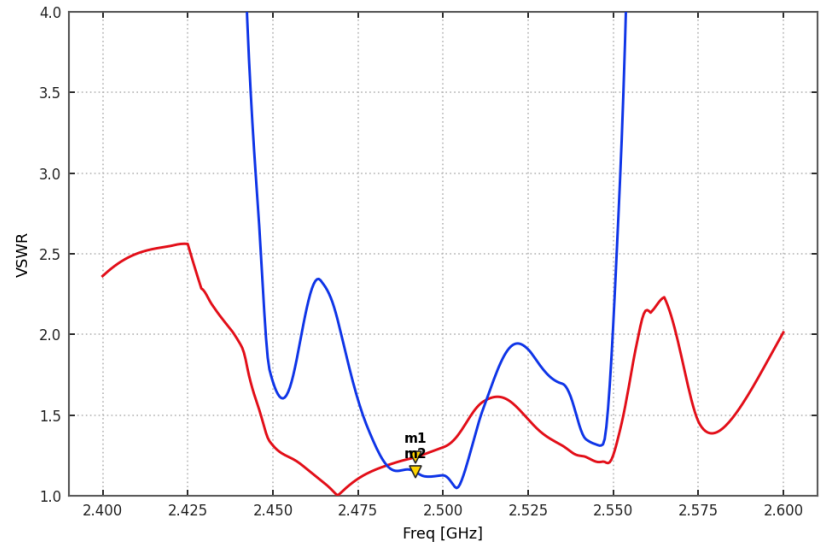
Curve Info

- dB(S(P2_S_OUT,P1_S_RHCP)) — SP_SBand : Sweep
- dB(S(P1_S_RHCP,P1_S_RHCP)) — SP_SBand : Sweep
- dB(S(P2_S_OUT,P2_S_OUT)) — SP_SBand : Sweep
- dB(S(P1_S_RHCP,P2_S_OUT)) — SP_SBand : Sweep

	Freq [GHz]	Value	Trace
m1	2.4920	34.954	S21
m2	2.4920	-19.393	S11
m3	2.4920	-23.280	S22
m4	2.4920	-58.184	S12
m5	2.3900	-55.874	S21
m6	2.5750	-72.437	S21

Figure - S-parameters.

XY Plot 6 - VSWR *SBand_2port*



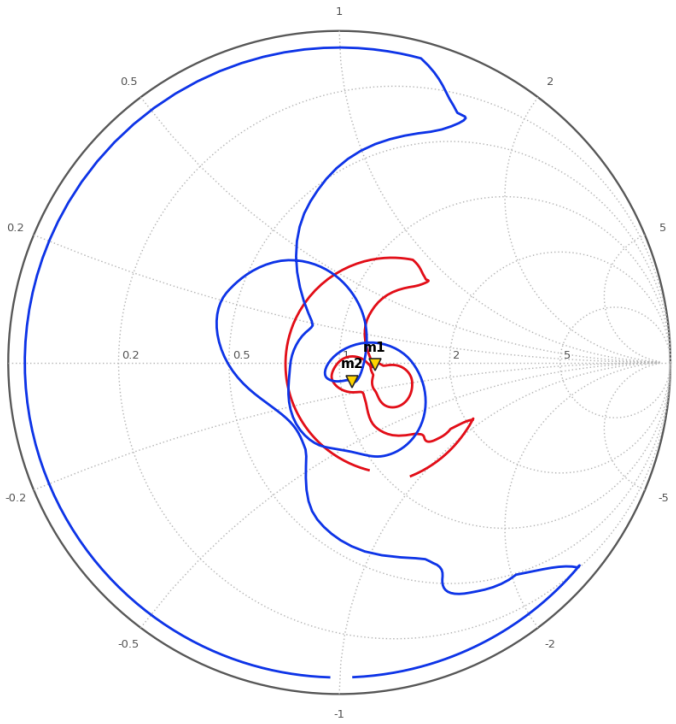
Curve Info

- VSWR(P1_S_RHCP) — SP_SBand : Sweep
- VSWR(P2_S_OUT) — SP_SBand : Sweep

	Freq [GHz]	Value	Trace
m1	2.4920	1.240	P1
m2	2.4920	1.147	P2

Figure - VSWR.

Smith Plot 1 - Input / output reflection *SBand_2port*



Curve Info

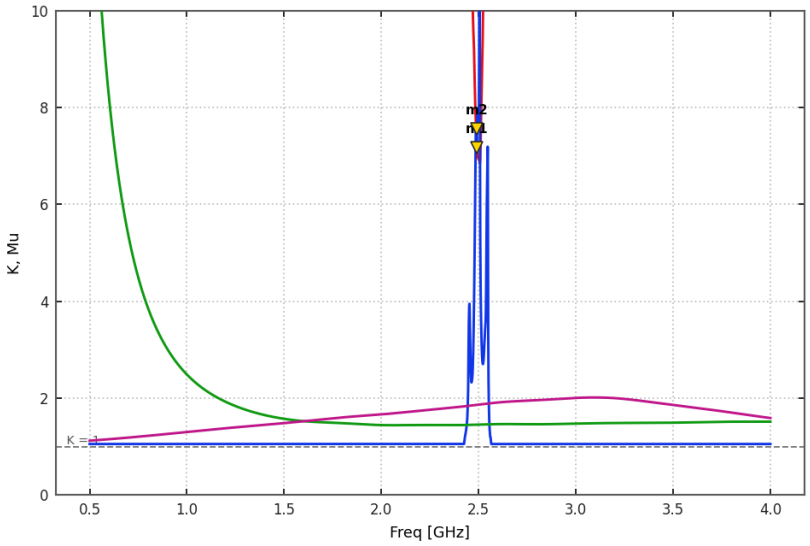
S(P1_S_RHCP,P1_S_RHCP)
SP_SBand : Sweep

S(P2_S_OUT,P2_S_OUT)
SP_SBand : Sweep

	Freq [GHz]	Value	Trace
m1	2.4920	62.0-j0.9	S11
m2	2.4920	53.6-j6.2	S22

Figure - Reflections.

XY Plot 7 - Stability *SBand_2port*



Curve Info

StabilityFactor K (system)
SP_SBand : Sweep

Mu1 (system)
SP_SBand : Sweep

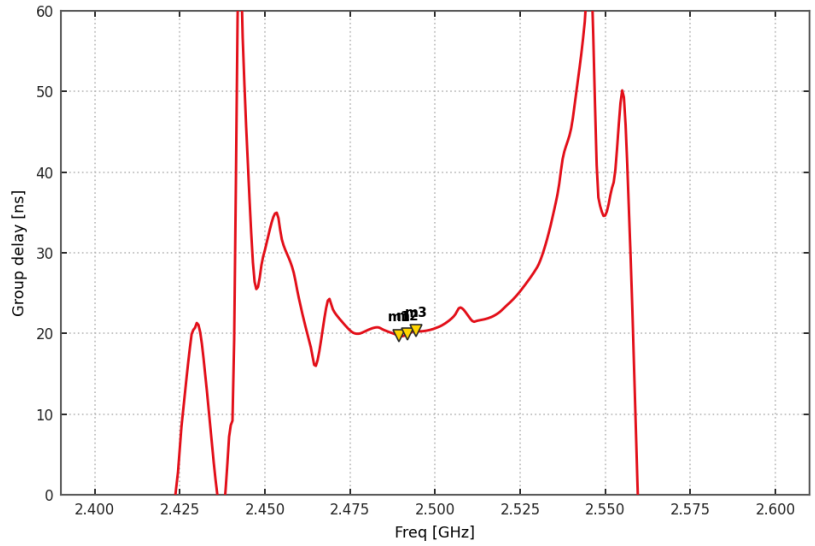
K (LNA stage, ref. design)
SP_SBand : Sweep

Mu1 (LNA stage)
SP_SBand : Sweep

	Freq [GHz]	Value	Trace
m1	2.4920	7.175	K
m2	2.4920	7.560	Mu1

Figure - Stability.

XY Plot 8 - Group delay *SBand_2port*



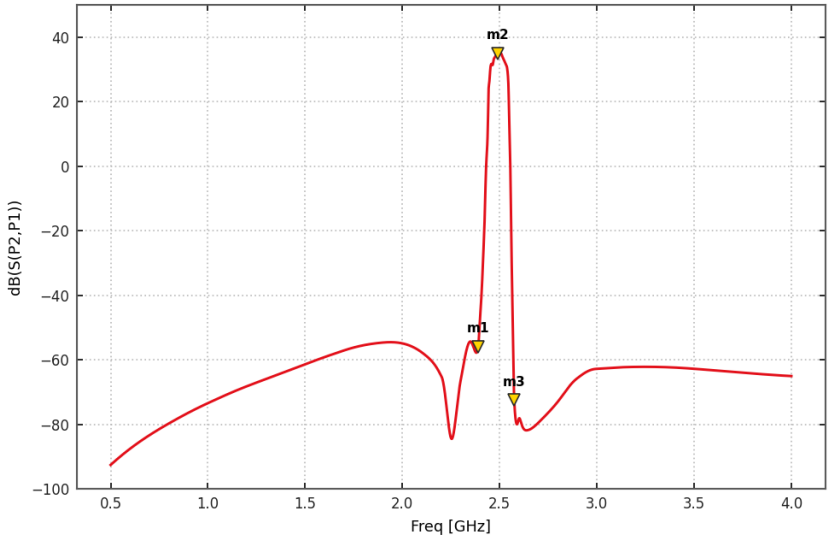
Curve Info

GroupDelay(P2_S_OUT,P1_S_RHCP) [ns]
SP_SBand : Sweep

	Freq [GHz]	Value	Trace
m1	2.4895	19.737	GD
m2	2.4920	19.927	GD
m3	2.4945	20.335	GD

Figure - Group delay (estimate).

XY Plot 3 - Wide-band transmission *SBand_2port*



Curve Info

dB(S(P2_S_OUT,P1_S_RHCP))
SP_SBand : Sweep

	Freq [GHz]	Value	Trace
m1	2.3900	-55.874	S21
m2	2.4920	34.954	S21
m3	2.5750	-72.437	S21

Figure - Wide-band transmission.

5 Filter rejection

5.1 L-band filters vs data-sheet limits

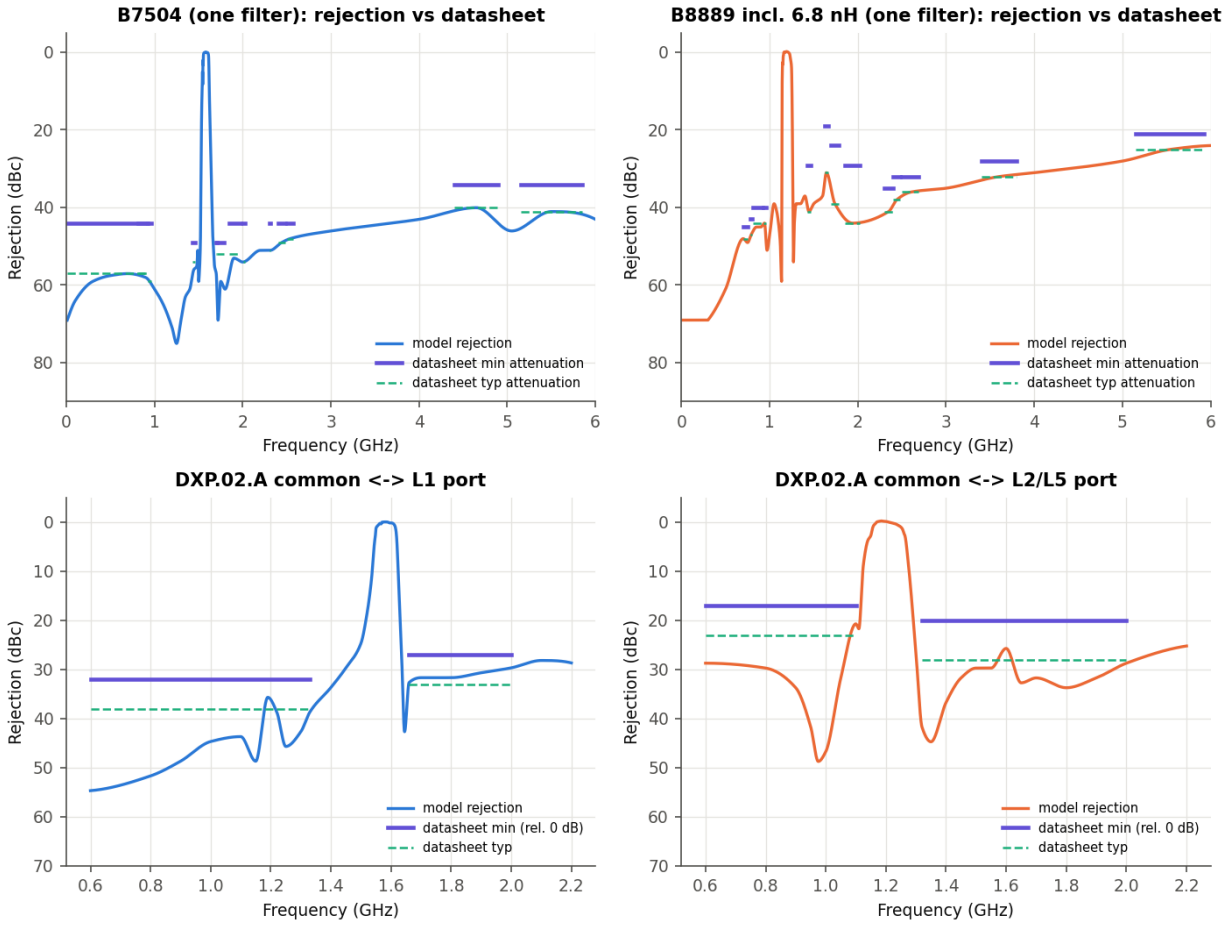


Figure - B7504, B8889 (+2 x 6.8 nH) and DXP.02.A paths vs data-sheet limits (all met).

Filter	Stop band (MHz)	Spec min dB	Spec typ dB	Model min dB	Margin dB
B7504	10-925	45	58	58.0	13.0
B7504	814-915	45	58	58.3	13.3
B7504	925-960	45	60	59.4	14.4
B7504	1427-1463	50	55	56.6	6.6
B7504	1695-1710	50	53	57.4	7.4
B7504	1710-1785	50	53	60.0	10.0
B7504	1850-1980	45	53	54.0	9.0
B7504	2010-2025	45	55	54.9	9.9
B7504	2305-2315	45	52	52.0	7.0
B7504	2401-2483	45	50	49.5	4.5
B7504	2500-2570	45	49	48.8	3.8
B7504	4400-4900	35	41	41.0	6.0
B7504	5150-5850	35	42	42.0	7.0
B8889	698-748	46	49	49.0	3.0
B8889	777-798	44	48	47.7	3.7
B8889	807-915	41	45	45.7	4.7
B8889	925-960	41	45	45.0	4.0
B8889	1427-1463	30	42	40.6	10.6
B8889	1626.5-1660.5	20	32	32.0	12.0
B8889	1695-1785	25	40	35.9	10.9
B8889	1850-2025	30	45	43.8	13.8
B8889	2300-2400	36	42	40.3	4.3

Filter	Stop band (MHz)	Spec min dB	Spec typ dB	Model min dB	Margin dB
B8889	2400-2483	33	39	38.2	5.2
B8889	2500-2690	33	37	36.7	3.7
B8889	3400-3800	29	33	32.5	3.5
B8889	5150-5925	22	26	25.1	3.1
DXP L1	600-1330	32	38	39.0	3.7
DXP L1	1660-2000	27	33	33.0	2.7
DXP L2/L5	600-1100	17	23	24.0	3.7
DXP L2/L5	1320-2000	20	28	29.0	5.7

Table 3 - L-band filter compliance (worst point in each specified stop band).

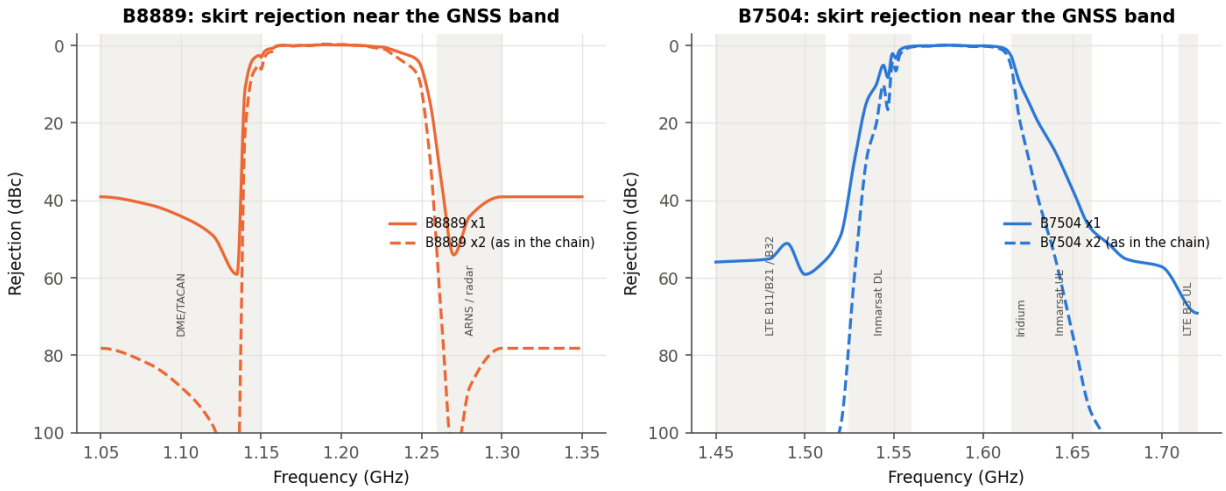
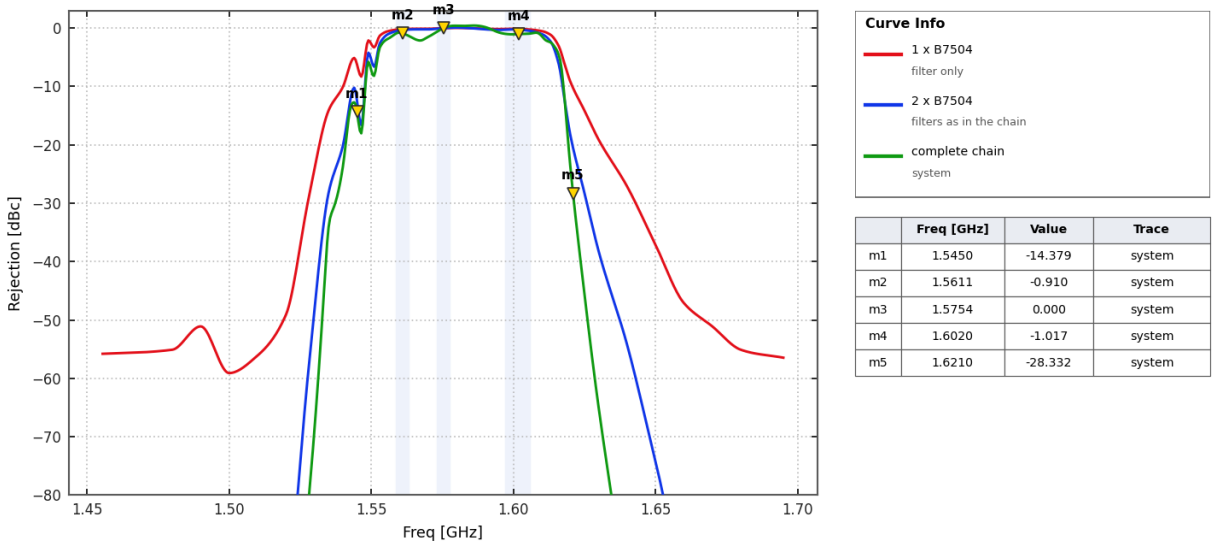


Figure - L-band SAW skirts next to the GNSS bands (one and two filters).

5.2 L1 chain

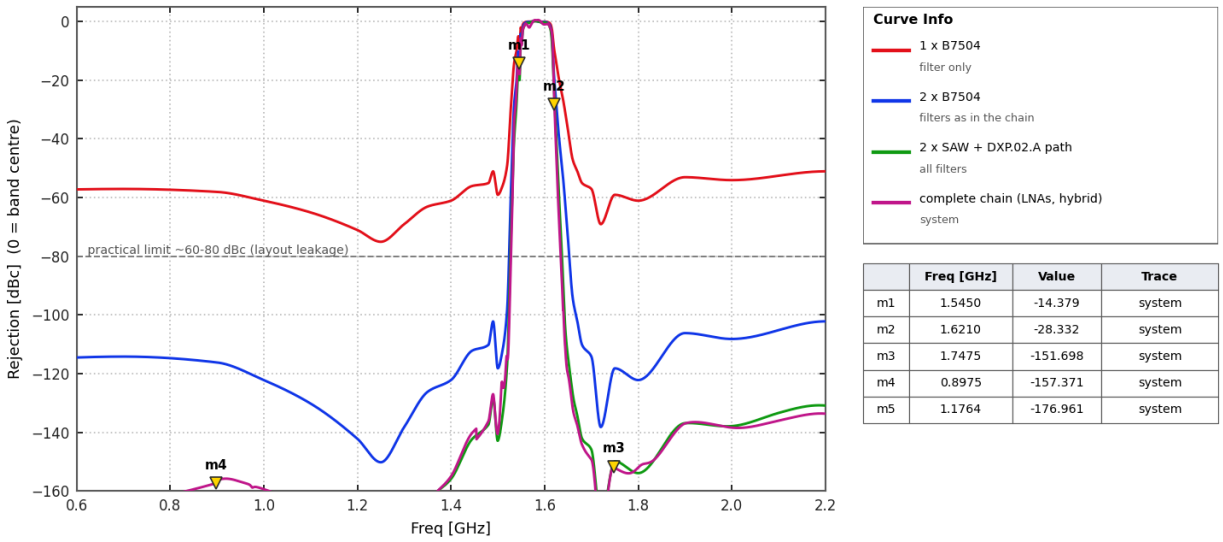
Rejection - L1 chain, in band and near band *ref. 1575.42 MHz*



	Freq [GHz]	Value	Trace
m1	1.5450	-14.379	system
m2	1.5611	-0.910	system
m3	1.5754	0.000	system
m4	1.6020	-1.017	system
m5	1.6210	-28.332	system

Figure - L1 chain: in band and near band (B1I, L1/E1, G1 shaded).

Rejection - L1 chain, out of band *ref. 1575.42 MHz*



	Freq [GHz]	Value	Trace
m1	1.5450	-14.379	system
m2	1.6210	-28.332	system
m3	1.7475	-151.698	system
m4	0.8975	-157.371	system
m5	1.1764	-176.961	system

Figure - L1 chain: out of band, per filter stage.

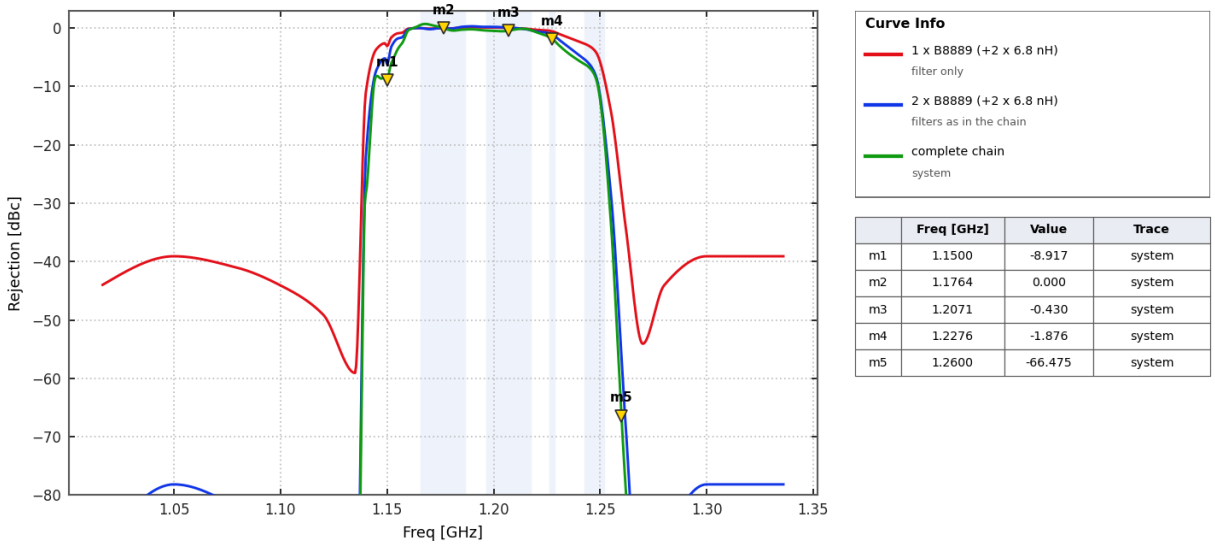
MHz	Service / point	1 SAW dBc	2 SAW dBc	2 SAW + DXP dBc	Complete chain dBc
787	LTE B13/B14 UL	57.3	114.6	166.6	161.9
897.5	GSM-900 / LTE B8 UL	58.1	116.2	164.9	157.4
1176.45	L5 (other branch)	69.5	139.0	178.2	177.0
1227.6	L2 (other branch)	73.8	147.6	188.0	192.5
1500	L-band (LTE B11/B21)	59.1	118.2	142.9	140.4
1530	Inmarsat DL	24.2	48.5	62.9	69.0
1545	Inmarsat DL	6.2	12.5	16.7	14.4
1550	lower skirt	2.7	5.4	6.7	7.2
1612	upper skirt	0.8	1.7	2.4	2.2
1616	Iridium low	3.1	6.2	7.8	4.7
1621	Iridium	10.3	20.5	26.4	28.3
1626.5	Iridium / Inmarsat UL	15.6	31.3	45.1	51.6
1643	Inmarsat UL	29.9	59.9	101.4	108.6

MHz	Service / point	1 SAW dBc	2 SAW dBc	2 SAW + DXP dBc	Complete chain dBc
1660	Inmarsat UL top	47.1	94.2	126.9	131.7
1710	LTE B3 UL low	63.5	127.1	158.8	162.0
1747.5	LTE B3 UL	59.3	118.6	150.3	151.7
1842.5	DCS-1800 DL	58.0	116.0	147.4	147.9
1950	LTE B1 UL	53.6	107.2	137.4	137.0
2140	LTE B1 DL	51.8	103.5	131.7	134.6
2442	Wi-Fi 2.4 GHz	49.1	98.2	-	-
5500	Wi-Fi 5 GHz	41.1	82.2	-	-

Table 4 - L1 chain rejection (dBc rel. 1575.42 MHz; '-' = outside model range; practical limit ~60-80 dBc).

5.3 L2/L5 chain

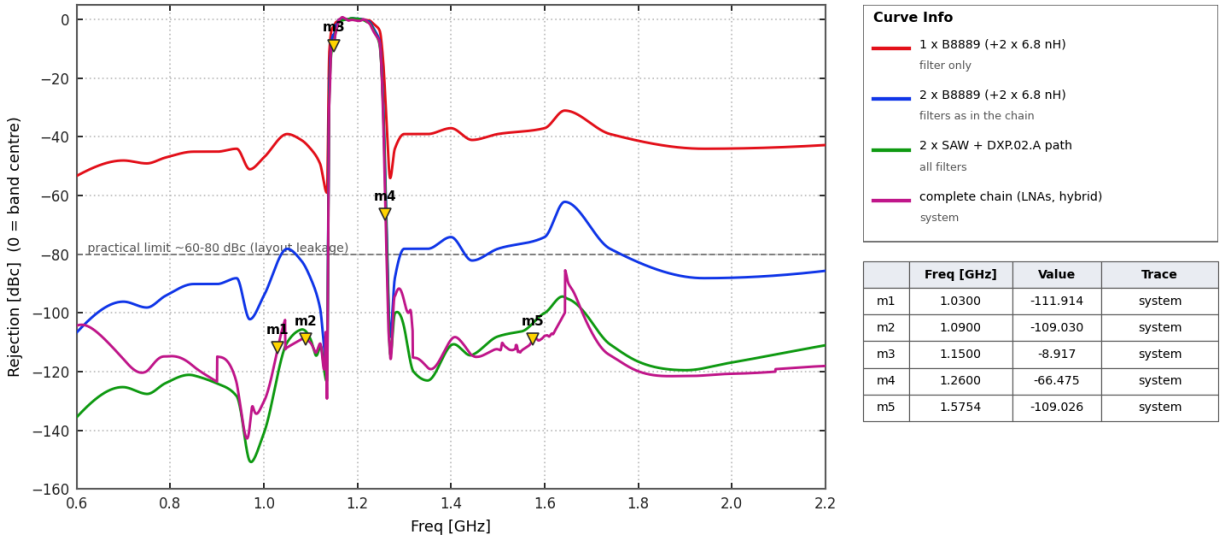
Rejection - L2/L5 chain, in band and near band *ref. 1176.45 MHz*



	Freq [GHz]	Value	Trace
m1	1.1500	-8.917	system
m2	1.1764	0.000	system
m3	1.2071	-0.430	system
m4	1.2276	-1.876	system
m5	1.2600	-66.475	system

Figure - L2/L5 chain: in band and near band (L5/E5a, E5b, L2, G2 shaded).

Rejection - L2/L5 chain, out of band *ref. 1176.45 MHz*



	Freq [GHz]	Value	Trace
m1	1.0300	-111.914	system
m2	1.0900	-109.030	system
m3	1.1500	-8.917	system
m4	1.2600	-66.475	system
m5	1.5754	-109.026	system

Figure - L2/L5 chain: out of band, per filter stage.

MHz	Service / point	1 SAW dBc	2 SAW dBc	2 SAW + DXP dBc	Complete chain dBc
787	LTE B13/B14 UL	47.2	94.5	124.2	114.8
897.5	GSM-900 / LTE B8 UL	45.1	90.2	123.9	123.2
960	DME/TACAN low	49.0	98.1	143.2	141.4
1030	SSR / IFF interrogation	41.2	82.4	120.8	111.9
1090	SSR / ADS-B reply	42.4	84.7	106.3	109.0
1150	DME top	3.1	6.2	9.0	8.9
1160	lower skirt	0.1	0.2	0.9	0.4
1250	upper skirt	5.8	11.6	12.8	12.3
1260	ARNS radar	28.2	56.4	58.8	66.5
1300	ARNS radar top	39.1	78.2	105.0	96.0
1400	L-band	37.1	74.2	111.0	109.2
1575.42	L1 (other branch)	37.8	75.6	103.4	109.0
1621	Iridium	34.0	68.0	96.5	106.4

MHz	Service / point	1 SAW dBc	2 SAW dBc	2 SAW + DXP dBc	Complete chain dBc
1710	LTE B3 UL low	36.7	73.4	105.3	109.2
1950	LTE B1 UL	44.1	88.2	118.5	121.2
2442	Wi-Fi 2.4 GHz	38.1	76.1	-	-
5500	Wi-Fi 5 GHz	25.2	50.5	-	-

Table 5 - L2/L5 chain rejection (dBc rel. 1176.45 MHz).

Path	At	Rejection (filters + DXP)
L1 branch	L5 1176.45 MHz	178 dBc
L1 branch	L2 1227.60 MHz	188 dBc
L2/L5 branch	L1 1575.42 MHz	103 dBc

Table 6 - Cross-band rejection between the L-band branches at the combined output (model).

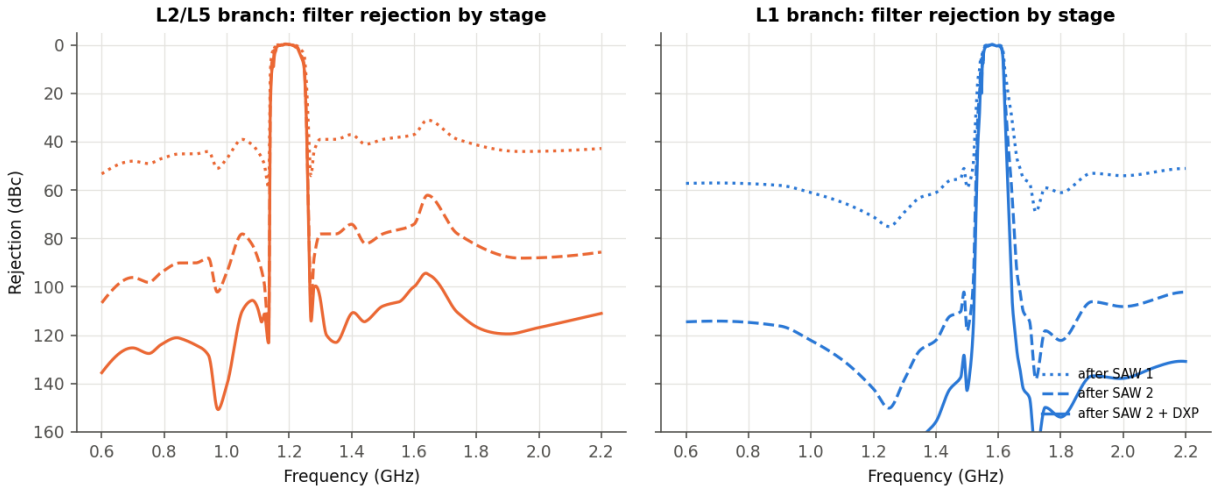


Figure - L-band cumulative filter rejection: SAW 1, SAW 2, SAW 2 + DXP.

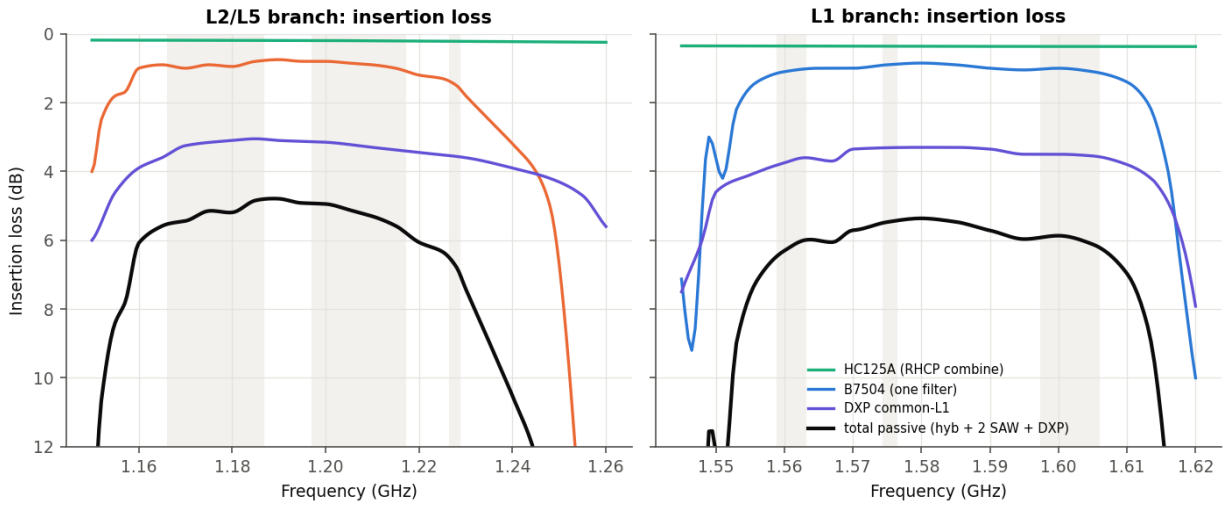


Figure - L-band in-band insertion loss: hybrid, one SAW, DXP path, total passive.

Chain	Band	1 SAW IL dB	2 SAW IL max dB	+DXP IL max dB	+DXP ripple dB	System ripple dB
L1	B1I	1.06	2.24	6.02	0.38	0.51
L1	L1 C/A / E1 / B1C	0.90	1.82	5.13	0.05	0.34
L1	L1 wide (E1/B1C 4.092 MHz)	0.90	1.86	5.18	0.13	0.79
L1	G1 (FDMA)	1.04	2.23	5.79	0.29	0.11
L2/L5	L5 / E5a / B2a	0.92	2.00	5.33	0.71	1.14
L2/L5	E5b / B2b	0.88	2.15	5.56	0.83	0.42
L2/L5	L2	1.48	3.12	6.69	0.32	0.41
L2/L5	G3 (CDMA)	0.85	1.88	5.22	0.48	0.41

Chain	Band	1 SAW IL dB	2 SAW IL max dB	+DXP IL max dB	+DXP ripple dB	System ripple dB
L2/L5	G2 (FDMA)	5.23	17.54	21.94	10.68	10.75

Table 7 - L-band in-band filter loss and ripple (G2 lies outside the B8889 pass-band).

5.4 NavIC S chain

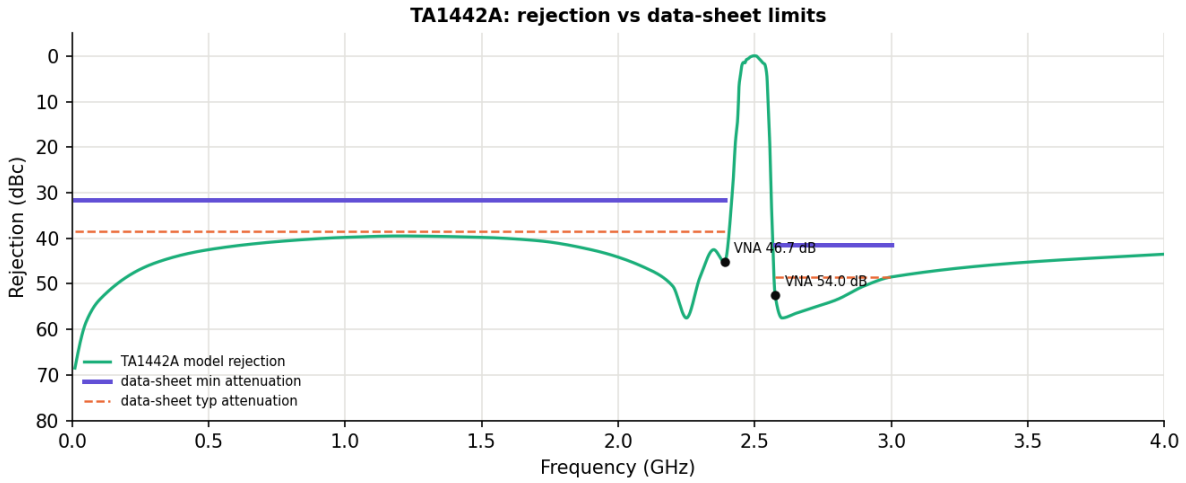
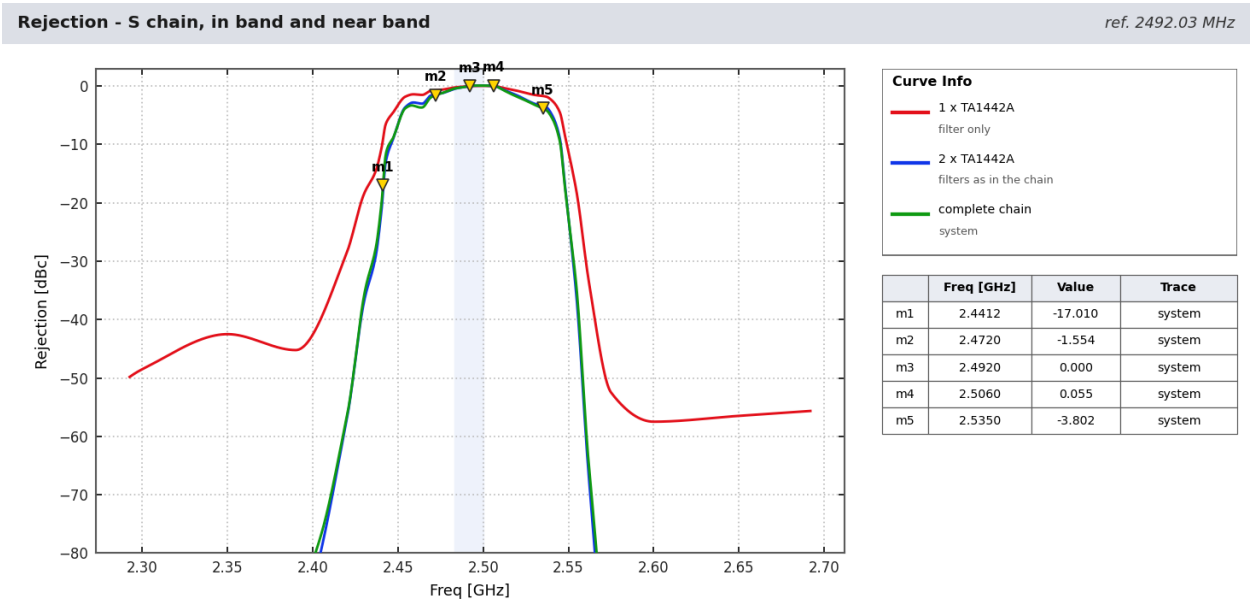


Figure - TA1442A vs data-sheet limits and VNA markers (met).



	Freq [GHz]	Value	Trace
m1	2.4412	-17.010	system
m2	2.4720	-1.554	system
m3	2.4920	0.000	system
m4	2.5060	0.055	system
m5	2.5350	-3.802	system

Figure - NavIC S chain: in band and near band.

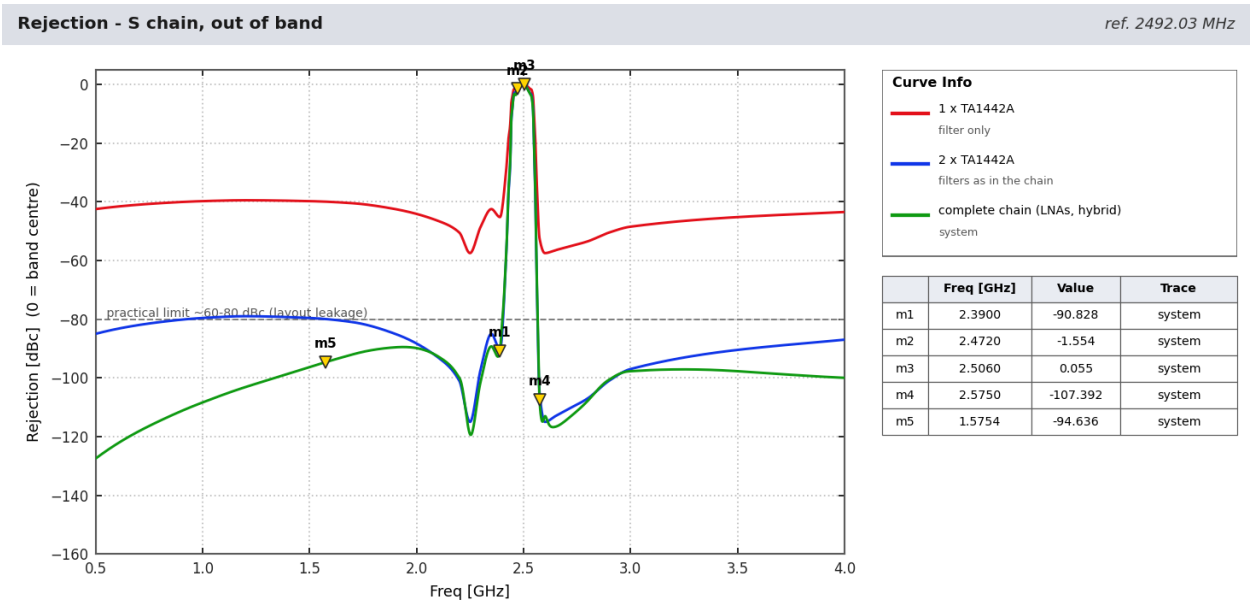


Figure - NavIC S chain: out of band, per filter stage.

MHz	Service / point	1 SAW dBc	2 SAW dBc	Complete chain dBc
1575.42	GNSS L1	40.0	80.0	94.6
2110	LTE B1 DL	46.8	93.6	93.1
2350	LTE B40	42.5	85.0	89.3
2390	TA1442A spec point	45.2	90.5	90.8
2412	Wi-Fi ch 1	34.7	69.3	68.2
2441	Bluetooth / ISM	9.5	19.0	17.8
2450	microwave oven	3.2	6.5	6.6
2472	Wi-Fi ch 13	0.7	1.4	1.6
2483.5	ISM top / Globalstar	0.2	0.4	0.4
2500	Globalstar top	0.0	0.0	0.0
2506	LTE B41 / n41	0.0	0.0	0.0
2535	LTE B7 UL	1.7	3.4	3.8
2570	LTE B7 UL top	47.6	95.2	92.3
2575	TA1442A spec point	52.5	105.0	107.4
2593	LTE B41 mid	57.1	114.2	114.5
2655	LTE B7 DL	56.4	112.8	116.5
3500	5G n78	45.2	90.5	97.7

Table 8 - NavIC S chain rejection (dBc rel. 2492.028 MHz).

Band	1 SAW IL dB	2 SAW IL max dB	System ripple dB
NavIC S SPS (2.046 MHz)	1.50	3.07	0.17
TA1442A spec band	1.50	3.12	0.22
NavIC S RS (16.5 MHz)	1.53	3.40	0.47

Table 9 - NavIC S in-band filter loss and ripple.

6 Interference

Chain	Interferer	MHz	At antenna dBm	LNA 2 out dBm	Status
L1	Iridium	1621	-20	12.0	near band - watch
L1	LTE B3 UL	1747.5	-10	-28.0	OK
L2/L5	DME / TACAN	1150	-30	13.3	RISK - above B8889 rating
L2/L5	GSM-900	897.5	-10	-6.6	OK
S	Wi-Fi ch 13	2472	-10	25.8	RISK - LNA 2 compressed
S	LTE B41	2506	-10	26.3	RISK - LNA 2 compressed
S	LTE B7 UL	2535	-10	24.5	RISK - LNA 2 compressed
S	5G n78	3500	-20	-35.8	OK

Table 10 - Interferer level at LNA 2 output (LNA OP1dB ~+20 dBm; B8889 +5 dBm, TA1442A +10 dBm max input).

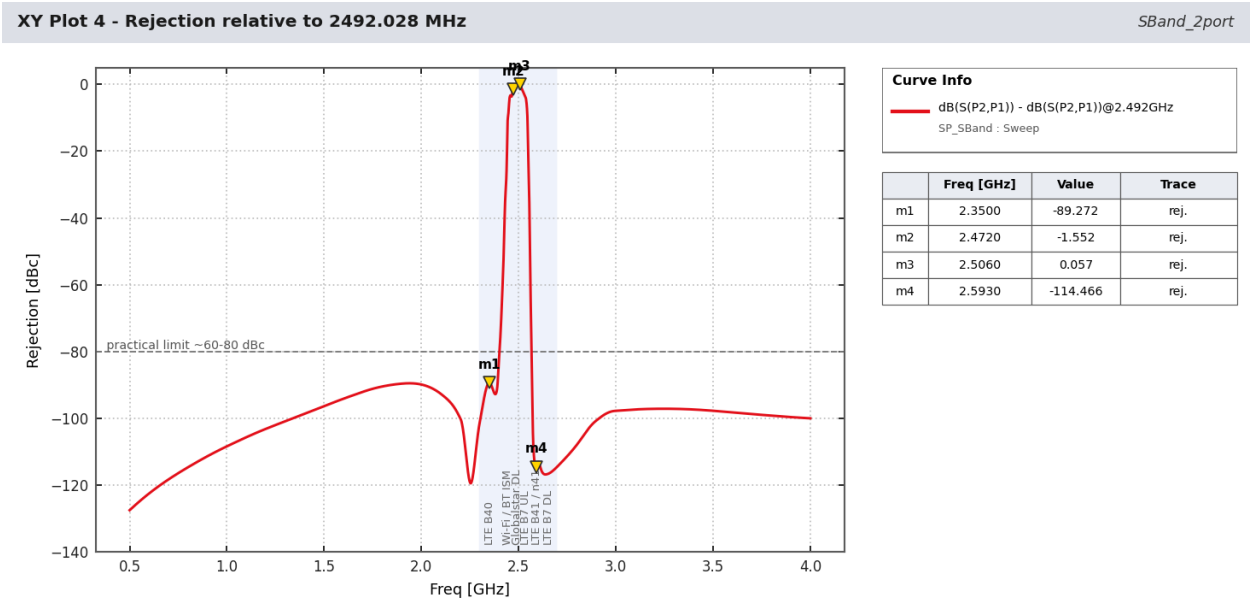


Figure - NavIC S chain rejection with the ISM / LTE bands.

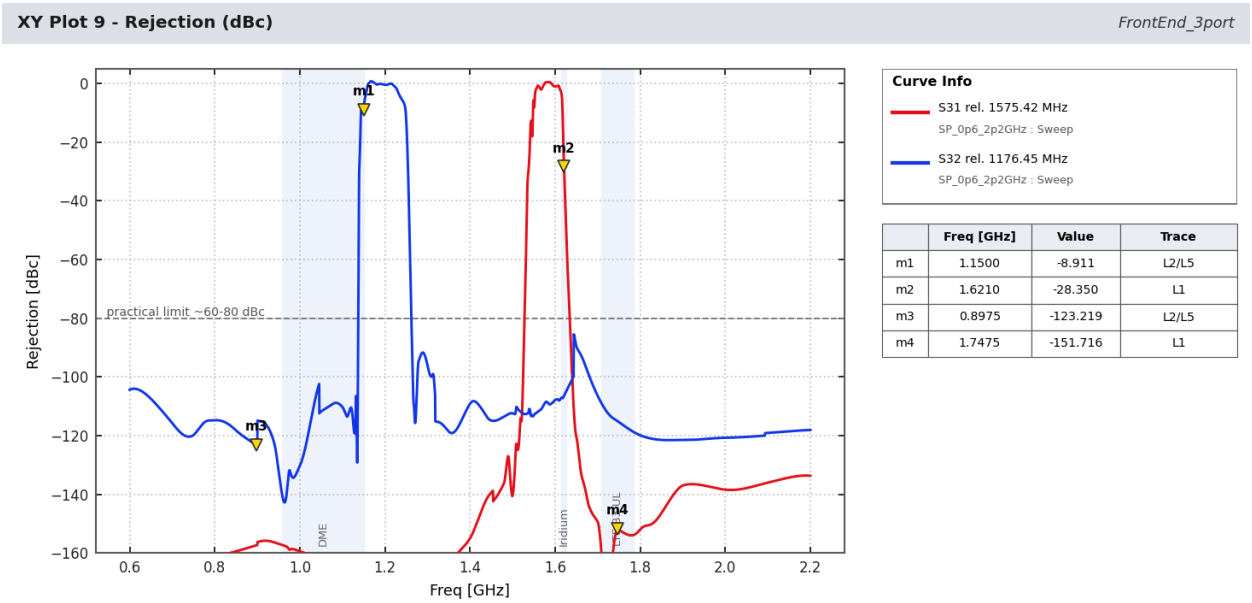


Figure - L-band chain rejection with DME / Iridium / LTE.

7 Cascade budgets

Stage	G dB	NF dB	Cum G	Cum NF	Cum IIP3	Cum IP1dB	Signal dBm	Noise dBm
HC125A (RHCP combine)	-0.36	0.36	-0.36	0.36	-	-	-128.9	-110.9
LNA1 TQP3M9037 + bias	21.95	0.37	21.59	0.73	13.2	0.3	-106.9	-88.5
B7504 #1	-0.89	0.89	20.70	0.74	13.2	0.3	-107.8	-89.4
LNA2 TQP3M9037 + bias	21.95	0.37	42.65	0.74	-7.9	-20.8	-85.8	-67.5
B7504 #2	-0.89	0.89	41.76	0.74	-7.9	-20.8	-86.7	-68.4
DXP.02.A (combiner)	-3.31	3.31	38.45	0.74	-7.9	-20.8	-90.0	-71.7

Cascade - L1 C/A (1575.42 MHz, 2.046 MHz).

Stage	G dB	NF dB	Cum G	Cum NF	Cum IIP3	Cum IP1dB	Signal dBm	Noise dBm
HC125A (RHCP combine)	-0.19	0.19	-0.19	0.19	-	-	-128.1	-100.9
LNA1 TQP3M9037 + bias	23.61	0.39	23.41	0.58	11.2	-0.9	-104.5	-76.9
B8889 #1	-0.91	0.91	22.50	0.59	11.2	-0.9	-105.4	-77.8
LNA2 TQP3M9037 + bias	23.61	0.39	46.11	0.59	-11.6	-23.6	-81.8	-54.2
B8889 #2	-0.91	0.91	45.20	0.59	-11.6	-23.6	-82.7	-55.1
DXP.02.A (combiner)	-3.14	3.14	42.06	0.59	-11.6	-23.6	-85.8	-58.2

Cascade - L5 (1176.45 MHz, 20.46 MHz).

Stage	G dB	NF dB	Cum G	Cum NF	Cum IIP3	Cum IP1dB	Signal dBm	Noise dBm
HC125A (RHCP combine)	-0.22	0.22	-0.22	0.22	-	-	-130.2	-110.9
LNA1 TQP3M9037 + bias	23.39	0.40	23.17	0.61	11.4	-0.7	-106.8	-87.1
B8889 #1	-1.45	1.45	21.71	0.62	11.4	-0.7	-108.3	-88.5
LNA2 TQP3M9037 + bias	23.39	0.40	45.10	0.62	-10.5	-22.7	-84.9	-65.1
B8889 #2	-1.45	1.45	43.65	0.62	-10.5	-22.7	-86.4	-66.6
DXP.02.A (combiner)	-3.56	3.56	40.09	0.62	-10.5	-22.7	-89.9	-70.2

Cascade - L2C (1227.60 MHz, 2.046 MHz).

Stage	G dB	NF dB	Cum G	Cum NF	Cum IIP3	Cum IP1dB	Signal dBm	Noise dBm
C2327J5003AHF (RHCP combine)	-0.43	0.43	-0.43	0.43	-	-	-132.7	-110.9
LNA1 TQP3M9037 (ref. design)	19.17	0.73	18.73	1.16	16.3	1.9	-113.6	-91.0
TA1442A #1	-1.50	1.50	17.23	1.18	16.3	1.9	-115.1	-92.5
LNA2 TQP3M9037 (ref. design)	19.17	0.73	36.40	1.19	-1.4	-15.9	-95.9	-73.3
TA1442A #2	-1.50	1.50	34.90	1.19	-1.4	-15.9	-97.4	-74.8

Cascade - NavIC S (2492.028 MHz, 2.046 MHz).

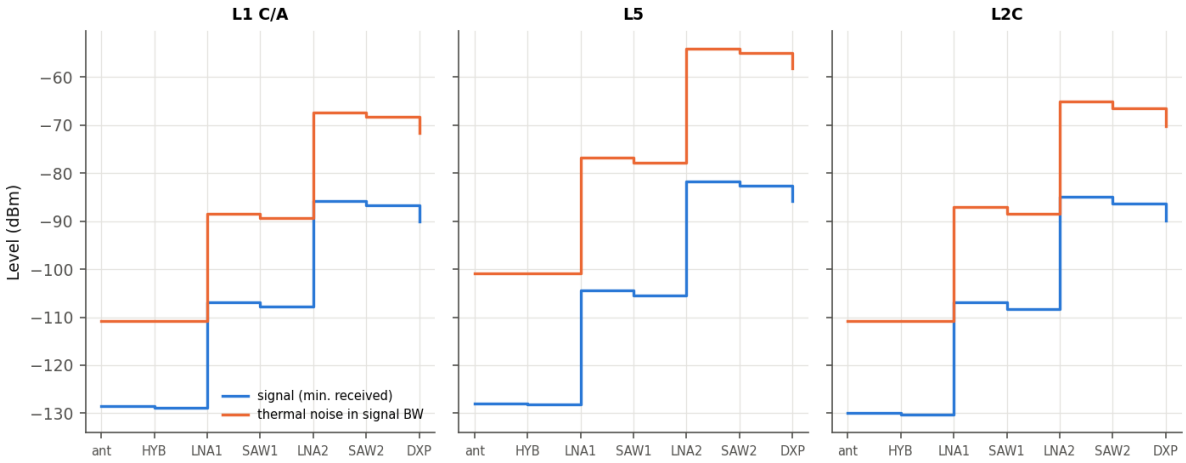


Figure - Level diagram (minimum signal and noise), L-band.

8 Actions

- NavIC S: narrow pre-filter (<= 20 MHz, > 30 dB at 2472 / 2506 MHz) before LNA 1.
- L2/L5: > 30 dB at 960-1150 MHz (DME) before LNA 1. L1: steeper upper skirt for Iridium.
- Reduce gain to 25-30 dB at the connector (one LNA per branch or a pad).
- Improve L1 / E5b input match (VSWR 2.3-2.75).
- Get measured S-parameters from the vendors; confirm TA1442A availability (2011 data sheet).
- Run the AEDT project on the licensed workstation; replace the ideal patch by HFSS data.

9 Parts

Part	Maker	Use	Qty
TQP3M9037	Qorvo	LNA, 0.4 dB NF, 5 V / 70 mA	6
HC125A	Taoglas	L-band 3 dB hybrid	2
B7504	Qualcomm RF360	L1 SAW	2
B8889	Qualcomm RF360	L2/L5 SAW (+2 x 6.8 nH)	2
DXP.02.A	Taoglas	L-band diplexer (combiner)	1
C2327J5003AHF	TTM / Xinger	S-band 3 dB hybrid	1
TA1442A	Tai-Saw	2492 MHz SAW	2
Patch elements	-	antenna (ideal in simulation)	3

Table 11 - Parts.